

Benefits of Diverse News Recommendations for Democracy

Juliane A. Lischka¹, Lucien Heitz^{2,3}, Alena Birrer⁴, Bibek Paudel⁵, Suzanne Tolmeijer², Laura Laugwitz¹, Abraham Bernstein^{2,3}

¹ Department of Journalism and Communication, University of Hamburg, Germany

² Department of Informatics, University of Zurich, Switzerland

³ Digital Society Initiative, University of Zurich, Switzerland

⁴ Department of Communication and Media Research, University of Zurich, Switzerland

⁵ Stanford University, USA

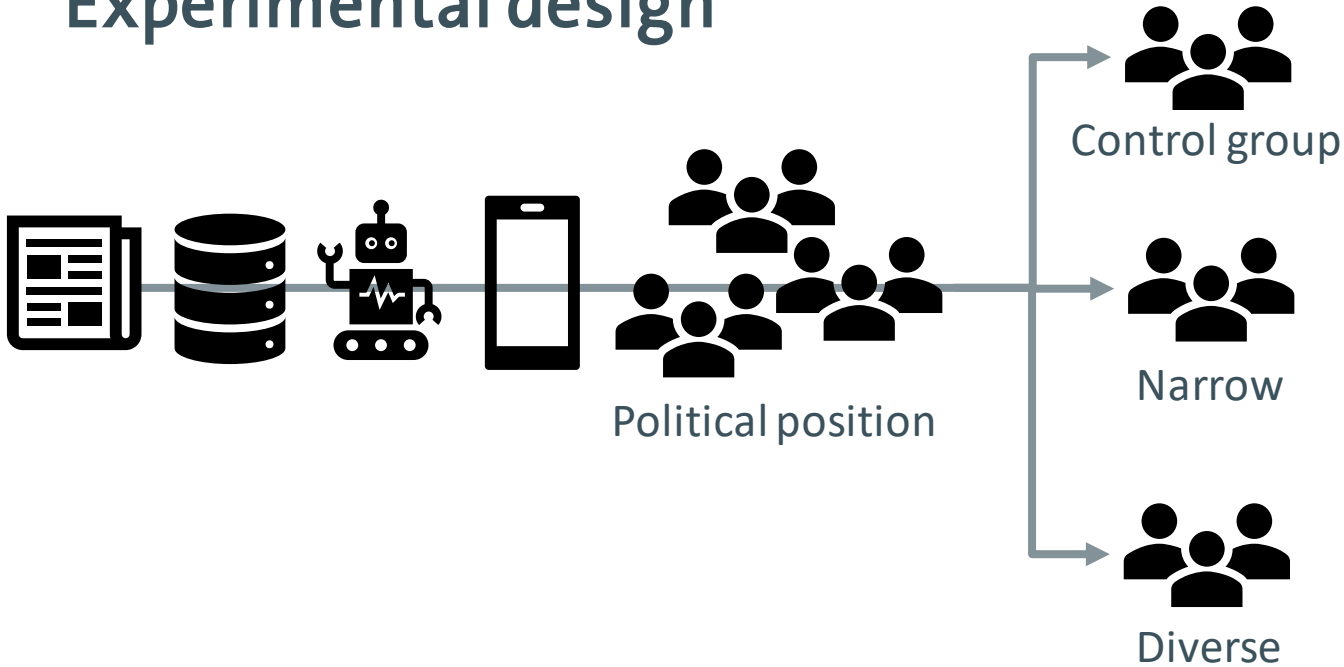
Algorithm and News session @ Future of Journalism 2021, 22. Sept 2021

News recommenders could

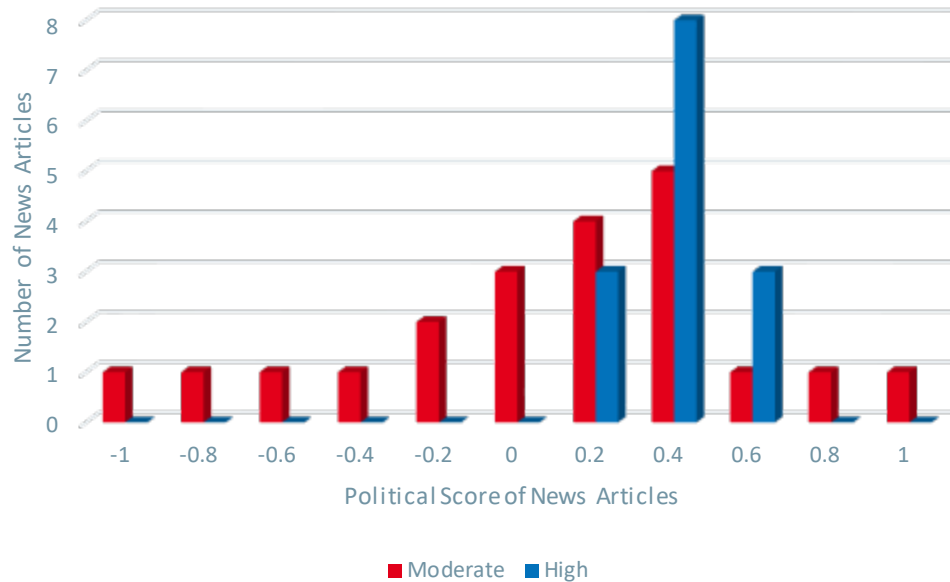
- a) create self-reinforcing biases that **damage** public debates (Milano et al., 2020)
- b) powerful tools to shape public opinion and serve as a foundation for **public cohesion** (Bernstein et al., 2021)

Background

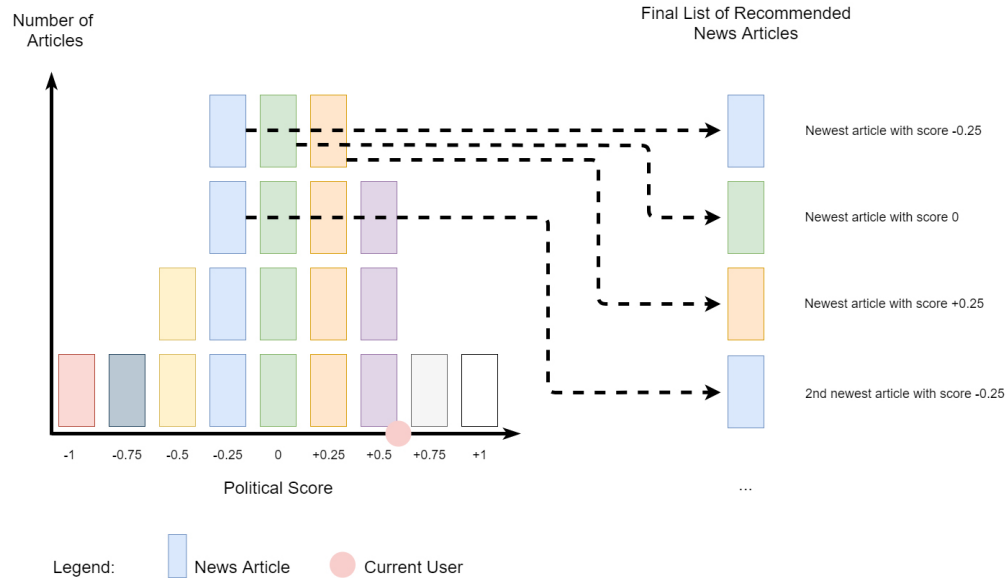
Experimental design



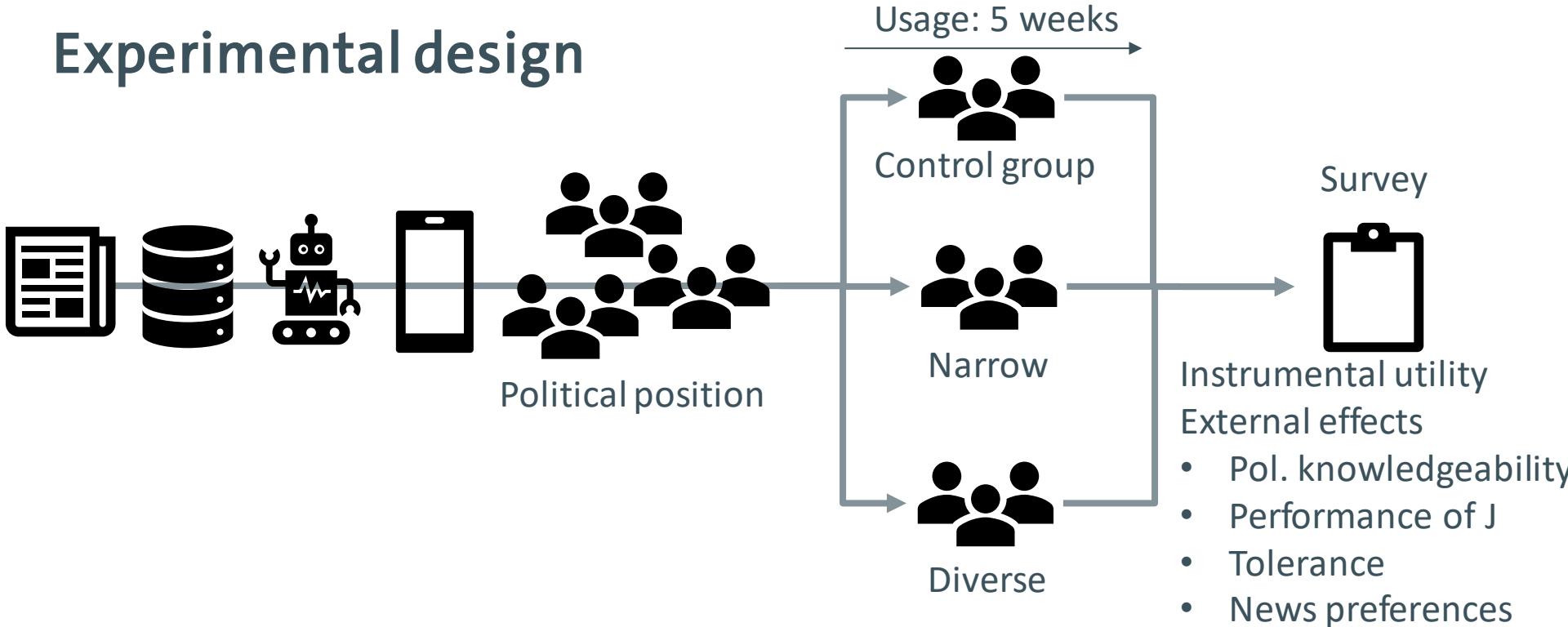
Narrow vs. diverse recommendations for a 0.4 user



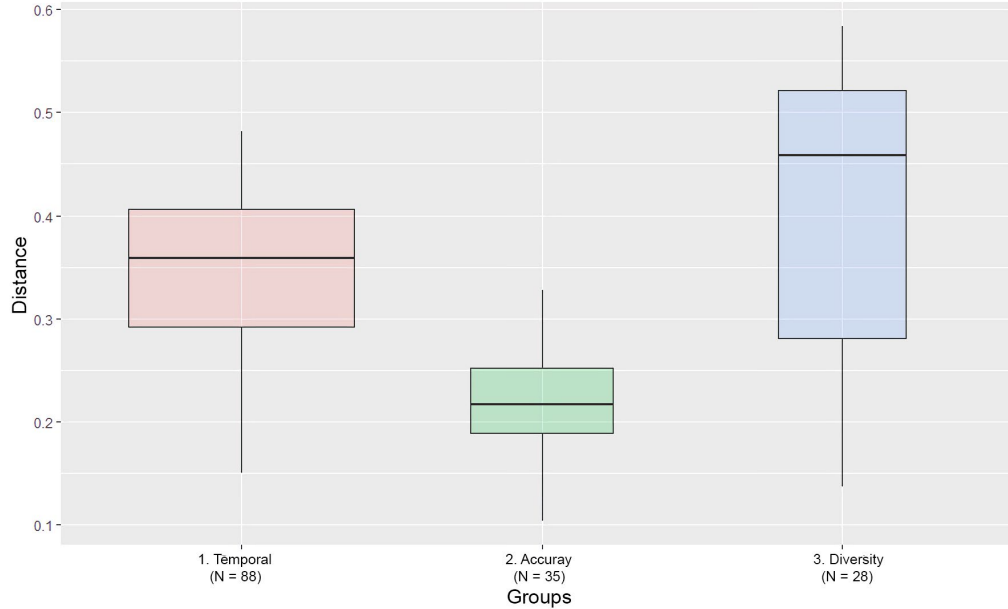
From article distribution to recommendation list



Experimental design



Article use and article distances



Survey results (1/2)

	Instrumental utility			Political interest			Political participation		
	β	ρ	η_p^2	β	ρ	η_p^2	β	ρ	η_p^2
Constant	3.668	0.000	0.468	1.227	0.004	0.057	2.242	0.000	0.233
Diversity Group	0.284	0.495	0.003	0.573	0.279	0.008	-0.136	0.754	0.001
Narrow Group	0.391	0.254	0.009	0.662	0.128	0.017	-0.144	0.686	0.001
Political position	0.037	0.900	0.000	0.270	0.476	0.004	-0.034	0.912	0.000
App usage (hrs)	0.020	0.009	0.048	-0.004	0.670	0.001	0.013	0.113	0.018
Ext. news usage	0.014	0.047	0.028	0.020	0.024	0.036	0.023	0.002	0.069
Gender (f = 1; m = 2)	-0.035	0.783	0.001	0.702	0.000	0.123	0.370	0.005	0.055
Age (y)	-0.005	0.241	0.010	-0.001	0.898	0.000	0.000	0.976	0.000
Education	-0.047	0.068	0.024	0.020	0.530	0.003	0.077	0.005	0.057
Diversity x Pol. pos.	-0.424	0.537	0.003	-0.855	0.329	0.007	-0.016	0.982	0.000
Narrow x Pol. pos.	-0.845	0.123	0.017	-0.558	0.421	0.005	0.185	0.745	0.001
$F(p)$	1.862 (.055)			4.467 (.000)			5.483 (.000)		
Adj. R^2	.055			.189			.231		
df	139, 10			139, 10			139, 10		

Survey results (2/2)

	Social performance of journalism			Tolerance for opposing views			News preferences for opposing views			News preferences for majority views		
	β	ρ	η_p^2	β	ρ	η_p^2	β	ρ	η_p^2	β	ρ	η_p^2
Cons.	1.465	0.000	0.087	2.869	0.000	0.401	2.937	0.002	0.068	3.770	0.000	0.090
Div.	1.098	0.032	0.033	1.443	0.000	0.097	0.903	0.054	0.026	0.748	0.159	0.014
Nar.	0.883	0.033	0.032	0.577	0.061	0.025	0.487	0.204	0.012	0.955	0.025	0.036
Pol. pos.	0.731	0.046	0.029	0.850	0.002	0.068	0.373	0.265	0.009	0.554	0.134	0.016
App use	0.012	0.195	0.012	0.013	0.062	0.025	0.022	0.009	0.049	0.017	0.060	0.025
Ext. news	0.009	0.263	0.009	0.018	0.005	0.055	0.026	0.000	0.088	0.013	0.095	0.020
Gender	0.140	0.361	0.006	-0.059	0.603	0.002	-0.065	0.647	0.002	-0.144	0.359	0.006
Age (y)	-0.003	0.574	0.002	-0.011	0.005	0.056	-0.553	0.506	0.003	-0.708	0.441	0.004
Edu.	0.100	0.002	0.070	0.068	0.004	0.059	0.008	0.768	0.001	-0.043	0.173	0.013
Div.xPos.	-1.628	0.053	0.027	-2.567	0.000	0.111	-1.200	0.122	0.017	-0.598	0.492	0.003
Nar.xPos.	-1.052	0.111	0.018	-0.813	0.098	0.020	-0.687	0.263	0.009	-1.289	0.058	0.026
$F(p)$	2.978 (.002)			4.971 (.000)			2.755 (.004)			1.635 (.103)		
Adj. R^2	.119			.210			.105			.041		
df	137, 10			139, 10			139, 10			139, 10		

1. Audiences find diverse news just as useful
 2. Diverse news may induce tolerance, especially for conservative users
 3. Diverse news can nudge users to prefer diverse news diet
- Diverse recommendations may have a de-polarizing capacity

Conclusion

Thank you!